

CHEKUNDA, Russian Federation

WMO: 315320

Lat: 50.8647N

Lon: 132.2580E

Elev: 272

StdP: 98.10

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.2	-39.4	-43.9	0.1	-41.0	-42.2	0.1	-39.1	5.3	-18.0	3.9	-24.2	0.1	0	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	29.9	20.1	27.9	19.4	26.0	18.8	22.2	26.6	21.2	25.5	20.3	24.0	1.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	16.2	24.0	19.9	15.1	22.8	19.0	14.3	22.1	66.9	26.4	63.2	25.7	59.9	24.0	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
5.5	5.0	3.7	DB	-42.8	32.9	2.0	1.8	-44.3	34.1	-45.4	35.1	-46.5	36.1	-47.9	37.4	
			WB	-42.9	24.2	1.8	1.4	-44.2	25.2	-45.3	26.0	-46.4	26.8	-47.7	27.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.2	-29.5	-23.1	-10.6	1.8	10.7	16.8	20.3	18.3	10.7	0.7	-15.1	-28.2	(6)
(7)		DBStd	18.38	5.22	5.73	6.23	4.48	3.78	3.42	2.63	3.04	4.05	4.59	8.45	6.03	(7)
(8)		HDD10.0	5345	1226	926	639	249	37	1	0	1	41	288	753	1184	(8)
(9)		HDD18.3	7613	1484	1159	897	497	238	68	11	37	229	546	1003	1443	(9)
(10)		CDD10.0	906	0	0	0	2	59	204	320	258	61	1	0	0	(10)
(11)		CDD18.3	133	0	0	0	0	1	22	73	37	1	0	0	0	(11)
(12)		CDH23.3	1143	0	0	0	1	48	284	564	239	7	0	0	0	(12)
(13)		CDH26.7	304	0	0	0	0	7	76	179	42	1	0	0	0	(13)
(14)	Wind	WSAvg	1.1	0.5	0.8	1.3	1.7	1.3	1.0	1.1	1.2	1.5	1.0	0.5	(14)	
(15)	Precipitation	PrecAvg	672	6	6	11	32	63	101	159	150	91	28	17	11	(15)
(16)		PrecMax	945	22	18	23	67	122	168	343	243	201	61	28	29	(16)
(17)		PrecMin	506	0	0	1	5	19	26	62	96	21	6	2	2	(17)
(18)		PrecStd	107	5	5	6	16	24	42	76	41	47	17	8	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-11.2	-4.6	7.6	20.4	28.0	31.7	33.2	30.0	24.1	15.9	4.0	-8.2	(19)
(20)			MCWB	-12.3	-6.7	2.4	10.2	15.6	22.2	21.0	20.1	18.0	9.0	1.5	-9.5	(20)
(21)		2%	DB	-16.6	-8.1	4.3	15.8	24.5	28.8	30.9	27.8	21.8	12.1	1.0	-12.7	(21)
(22)			MCWB	-17.1	-9.7	0.2	7.8	13.3	18.4	20.5	20.5	15.8	6.8	-0.6	-13.2	(22)
(23)		5%	DB	-19.5	-11.0	1.8	12.6	22.0	26.6	28.7	26.0	19.7	9.7	-0.8	-16.1	(23)
(24)			MCWB	-20.0	-12.2	-1.4	5.7	12.3	17.6	20.1	19.5	14.5	5.5	-2.0	-16.6	(24)
(25)		10%	DB	-21.9	-13.7	-0.5	9.8	19.3	24.4	26.5	24.3	17.7	7.4	-3.3	-19.4	(25)
(26)			MCWB	-22.3	-14.6	-3.2	4.0	10.9	17.1	19.8	18.8	13.4	4.2	-4.4	-19.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-12.3	-6.4	2.7	11.4	16.7	23.6	24.1	23.4	19.3	10.7	1.6	-9.4	(27)
(28)			MCDWB	-11.6	-4.8	6.5	19.4	24.1	29.0	28.6	26.8	23.1	14.3	3.2	-8.5	(28)
(29)		2%	WB	-17.0	-9.6	0.6	8.6	14.6	21.1	22.4	21.8	17.1	7.9	-0.3	-13.2	(29)
(30)			MCDWB	-16.6	-8.1	3.7	14.3	21.4	26.8	26.9	25.4	20.4	11.1	0.8	-12.8	(30)
(31)		5%	WB	-19.9	-12.1	-1.3	6.3	13.3	19.2	21.5	20.8	15.7	6.2	-2.1	-16.5	(31)
(32)			MCDWB	-19.6	-11.0	1.7	11.5	19.4	24.0	26.0	24.1	18.5	8.6	-0.7	-16.1	(32)
(33)		10%	WB	-22.2	-14.6	-3.2	4.5	12.0	17.9	20.7	19.8	14.3	4.6	-4.4	-19.8	(33)
(34)			MCDWB	-21.9	-13.6	-0.6	9.3	17.7	22.4	24.9	22.8	16.8	6.9	-3.3	-19.4	(34)
(35)	Mean Daily Temperature Range	MDBR	10.4	13.9	14.3	12.4	13.4	12.3	11.1	10.7	11.5	9.8	9.1	8.9	(35)	
(36)		5% DB	MCDBR	10.1	14.2	14.2	17.2	19.0	16.6	15.5	13.5	14.1	13.9	8.9	8.8	(36)
(37)			MCWBR	9.5	12.7	10.4	9.9	9.0	7.5	6.4	6.6	7.9	9.0	7.2	8.2	(37)
(38)			5% WB	MCDWB	10.0	14.0	13.3	16.0	15.8	14.3	11.8	10.6	12.0	11.5	8.6	8.5
(39)		MCWBR		9.4	12.5	9.7	9.7	8.3	7.6	6.1	6.1	7.5	8.2	7.1	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.264	0.298	0.365	0.493	0.559	0.511	0.486	0.416	0.386	0.394	0.326	0.268	(40)	
(41)		taud	2.152	2.062	1.934	1.791	1.765	1.981	2.109	2.312	2.309	2.093	2.091	2.110	(41)	
(42)		Ebn at Noon	749	811	810	741	712	757	772	814	797	683	640	668	(42)	
(43)		Edh at Noon	74	108	150	196	211	171	149	116	104	106	80	66	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.63	2.81	4.39	4.98	4.97	5.53	5.11	4.52	3.46	2.22	1.53	1.26	(44)	
(45)		RadStd	0.04	0.07	0.19	0.51	0.47	0.65	0.40	0.37	0.35	0.16	0.12	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	+0.47	+0.61	N/A	N/A	N/A	N/A

Nomenclature: See separate page